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Question 1

Not answered

Marked out of 1.00

Question ID #10633: Autocorrelation would most likely be a problem when an investigator uses which of the following research designs?

Select one:

- ☐ A. Between group
- ☐ B. Latin square
- ☐ C. Solomon four-group
- ☐ D. Time-series

The correct answer is D.

Autocorrelation is a problem whenever observations are not independent. Of the research designs listed, only the time-series design involves non-independent observations (i.e., it involves measuring the same people repeatedly before and after applying an intervention).

Answer A: Autocorrelation is a problem whenever observations are not independent. A between group design is used for independent observations and is used to compare outcomes from different groups instead of measuring the same people repeatedly before and after applying an intervention.

Answer B: A Latin Square design is used to control or allow two extraneous sources of variability while investigating a third and does not involve non-independent observations.

Answer C: The Solomon four-group design is a method for addressing concerns that the treatment was sensitized by a pretest and does not involve non-independent observations.

Question 2

Not answered

Marked out of 1.00

Question ID #10712: For their dissertation research, a graduate student administered a measure of state anxiety to a group of college students on five consecutive days before and after the students participated in a stress reduction workshop. Which of the following research designs did the graduate student use?

Select one:

- ☐ A. Time-series
- ☐ B. Latin square
- ☐ C. Static-group comparison
- ☐ D. Multiple-baseline

The correct answer is A.

The time-series design is a type of within-subjects design that involves evaluating the effects of an intervention by comparing multiple quantitative observations of participants before and after they are exposed to the intervention.

Answer B: The Latin square design is a type of counterbalanced design used to control carryover effects.

Answer C: The static-group comparison design includes two intact groups, one that receives the treatment and one that does not, and involves evaluating the effects of the treatment by comparing the post-test scores of the two groups. In this case, all students participated in the intervention (the stress reduction workshop) and completed measures of state anxiety.

Answer D: The multiple-baseline design is a single-case design that involves comparing the effects of an intervention across two or more settings, behaviors, or participants.

Question 3

Not answered

Marked out of 1.00

Question ID #11061: Which of the following would be useful for measuring the strength of the relationship between two continuous variables when their relationship is nonlinear?

Select one:

- ☐ A. Eta coefficient
- ☐ B. Phi coefficient
- ☐ C. Spearman rho
- ☐ D. Pearson r

The correct answer is A.

Eta is also known as the eta coefficient and the correlation ratio can be used when the relationship between two continuous variables is nonlinear.

Answer B: The phi coefficient is used when both variables are true dichotomies.

Answer C: Spearman rho is used when both variables are ranks.

Answer D: The Pearson r is used when both variables are continuous and their relationship is linear.

Question 4

Not answered

Marked out of 1.00

Question ID #11696: When using protocol analysis, a researcher asks an individual to ____.

Select one:

- ☐ A. brainstorm with other problem-solvers
- ☐ B. think aloud while solving a complex problem
- ☐ C. identify alternative strategies for solving the problem
- ☐ D. construct a cognitive map of a problem

The correct answer is B.

Protocol analysis is a method for obtaining information about a person's thoughts while solving problems. It involves having the individual "think aloud" while solving a problem and an observer records what the individual says; the observer then analyzes the individual's verbalizations in order to identify the cognitive processes involved in solving the problem.

Answer A: Researchers use protocol analysis to understand an individual's thought processes as they solve a problem. Brainstorming with others would not be considered an application of protocol analysis as it involves a dynamic process of generating ideas with others.

Answer C: Researchers use protocol analysis to identify cognitive processes involved in solving a problem. While a researcher may observe an individual identify alternative solutions to a problem, this is not the best answer because a researcher would not directly ask the individual to engage in this specific strategy, and the researcher would not gain insight into how the individual arrived at their responses which is the goal of protocol analysis.

Answer D: Researchers use protocol analysis to identify cognitive processes involved in solving a problem. While a researcher may observe an individual construct a cognitive map, this would not be the best answer because a researcher would not directly ask the individual to engage in this specific strategy.

Question 5

Not answered

Marked out of 1.00

Question ID #11707: When reviewing the data collected from their study's three treatment groups, a researcher discovers that the standard deviations of the three groups differ. If the researcher analyzes the data using the t-test for independent samples, this is ____.

Select one:

- ☐ A. not a problem
- ☐ B. not likely to be a problem as long as the three groups have the same number of participants
- ☐ C. not likely to be a problem as long as all three distributions have a rectangular shape
- ☐ D. not likely to be a problem as long as the data represent an interval or ratio scale of measurement

The correct answer is B.

When groups are equal in size, a violation of the assumption of homogeneity of variances will probably not substantially affect the results of the t-test.

Answer A: To consider this situation as "not a problem," the three groups must have the same number of participants. This is not the best response as the required contextual information is missing.

Answer C: One of the assumptions of the t-test is that the distribution of scores is normally shaped, not rectangular.

Answer D: Use of the t-test already requires that the data represent an interval or ratio scale.

Question 6

Not answered

Marked out of 1.00

Question ID #11709: An investigator uses a factorial ANOVA to assess the effects of two independent variables on a dependent variable and obtains significant main and interaction effects. When interpreting the results, the investigator should ____.

Select one:

- ☐ A. ignore the interaction since the main effects are significant
- ☐ B. interpret the main effects with caution since the interaction is significant
- ☐ C. interpret the interaction with caution since the main effects are significant
- ☐ D. re-analyze the data since these results are not possible

The correct answer is B.

Main effects refer to the effects of one independent variable on the dependent variable when considered alone, while interaction effects refer to the effects of one independent variable at different levels of another independent variable. When the interaction is significant, the effects of one independent variable differ for different levels of another independent variable and it is not possible to conclude that the independent variable has consistent main effects.

Answer A: A researcher should not ignore significant interaction effects regardless of whether the main effect is significant. When there are significant interaction effects, the researcher should actually interpret the main effect with caution. When the interaction is significant, the effects of one independent variable differ for different levels of another independent variable and it is not possible to conclude that the independent variable has consistent main effects.

Answer C: Significant main effects do not dictate the interpretation of significant interaction effects. A researcher should interpret the main effect with caution and focus on the significant interaction effect because the effects of one independent variable differ for different levels of another independent variable and it is not possible to conclude that the independent variable has consistent main effects.

Answer D: It is possible to observe any combination of significant main and/or interaction effects. How the researcher should interpret these results is the bigger question.

Question 7

Not answered

Marked out of 1.00

Question ID #11711: To analyze the data collected in a study that includes a single independent variable and three dependent variables, we would use a MANOVA (multivariate analysis of variance) rather than three separate one-way ANOVAs because ____.

Select one:

- ☐ A. we want to statistically remove the effects of a confounding variable
- ☐ B. at least one of the dependent variables is an extraneous variable
- ☐ C. we want to reduce the probability of making a Type I error
- ☐ D. we want to avoid conducting a post-hoc analysis

The correct answer is C.

Researchers use the MANOVA to simultaneously assess the effects of one or more independent variables on two or more dependent variables. If three separate ANOVAs were conducted at a level of significance of .05 each, the "experiment wise error rate" (probability of making a Type I error) would be increased to approximately .15. By conducting a single MANOVA at the .05 level, the experiment wise error rate remains at .05.

Answer A: The analysis of covariance (ANCOVA) is used to statistically remove the effects of an extraneous variable on the dependent variable.

Answer B: Extraneous variables are factors in an experiment that are not being studied and these are not related to the question.

Answer D: If the MANOVA yields significant results, the researcher will have to conduct post hoc tests (probably three separate ANOVAs) to determine if the independent variable affected all or only some of the dependent variables.

Question 8

Not answered

Marked out of 1.00

Question ID #11715: Squaring the multiple correlation coefficient (R) produces a measure of the ____.

Select one:

- ☐ A. relative amount of dispersion (variability) in a set of scores
- ☐ B. degree to which a true score can be expected to vary from an obtained score
- ☐ C. correlation between two variables when the effects of a third variable have been statistically removed
- ☐ D. the amount of variability shared by three or more variables

The correct answer is D.

We use the multiple correlation coefficient to measure the degree of association among three or more variables. Like any other correlation coefficient, we can square it to obtain a measure of shared variability.

Answer A: We use the coefficient of variation to provide this information -- it is an infrequently used measure of variability.

Answer B: This describes the standard error of measurement.

Answer C: This describes first-order correlation.

Question 9

Not answered

Marked out of 1.00

Question ID #11716: A researcher would use an ABAB design rather than an AB design in order to control which of the following threats to the study's validity?

Select one:

- ☐ A. Diffusion
- ☐ B. Attrition
- ☐ C. Instrumentation
- ☐ D. History

The correct answer is D.

An ABAB design is a single-subject design that involves collecting baseline data, administering the independent variable, removing the independent variable, and then readministering the independent variable. The use of an ABAB design helps determine whether or not an observed change in the dependent variable is due to the independent variable or to an external event (history). Specifically, if the participant exhibits the same change after the independent variable is administered the second time that they exhibited after it was administered the first time, that change can be assumed to have been caused by the independent variable rather than by history.

Answer A: Diffusion is a threat to a study's validity when participants in one group (often a no-treatment control group) benefit from the intervention administered to another group (e.g., when participants in the control group inadvertently learn about the treatment or are accidentally exposed to it).

Answer B: Attrition is a threat to a study's internal validity. It occurs when participants who drop out of one group differ in a systematic way from those who drop out of another group and this difference affects the study's results. The ABAB design would not be used in this case because we are making between-groups comparisons and the ABAB design is used for within groups comparisons or single-subject design.

Answer C: Instrumentation is a threat to internal validity and refers to any change in tests or other measuring devices administered to participants during the course of the study that confounds the study's results. An ABAB design would not help control instrumentation effects.

Question 10

Not answered

Marked out of 1.00

Question ID #11718: A research participant's score on the dependent variable is the amount of time (minutes) it took to complete a task. When assigning scores to participants, the researcher discovers that three of the 60 participants did not complete the task, and the researcher assigns them the maximum amount of time given to participants to work on the task. Which of the following is the best measure of central tendency for the data collected in this study?

Select one:

- ☐ A. Harmonic mean
- ☐ B. Arithmetic mean
- ☐ C. Mode
- ☐ D. Median

The correct answer is D.

The mean is affected by the magnitude of every score in a distribution, but the median is not. In this situation, the median would not be affected by the magnitude of the three missing scores and, therefore, would be a more accurate measure of central tendency.

Answer A: The harmonic mean is an average and is adversely affected by missing data.

Answer B: Ordinarily, the best measure of central tendency for interval or ratio data is the arithmetic mean, but it is also more impacted by missing data. In the situation described in this question, the mean would not be the most accurate measure because the distribution of data includes three scores that are estimates (rather than actual measures) of the value of interest.

Answer C: The mode provides less information about a distribution than the median (i.e., the mode only indicates the most frequently occurring score in the distribution), so the median would be a better measure of central tendency.

Question 11

Not answered

Marked out of 1.00

Question ID #11719: When the correlation between high school students' GPA and amount of time they spend studying each week is ____, this means that about 15% of variability in GPA is accounted for by the amount of time spent studying.

Select one:

- ☐ A. 0.02
- ☐ B. 0.15
- ☐ C. 0.39
- ☐ D. 0.85

The correct answer is C.

To determine the amount of variability in one variable that is explained by variability in another variable (i.e., the amount of shared variability), you must square the correlation coefficient. Conversely, to determine the correlation coefficient when the shared variability is known, you must calculate the square root of the shared variability. The square root of .15 is .387, so .39 is the best answer of the numbers listed.

Answers A, B, and D: To determine the correlation coefficient when the shared variability is known, you must calculate the square root of the shared variability. The square root of .15 is .387, so .39 is the best answer of the numbers listed.

Question 12

Not answered

Marked out of 1.00

Question ID #11721: In which of the following types of research do research participants act as their own no-treatment "controls"?

Select one:

- ☐ A. Single-subject
- ☐ B. Cross-sectional
- ☐ C. Ex post facto
- ☐ D. Double-blind

The correct answer is A.

In research, a control group is a comparison group that does not receive treatment or, alternatively, receives a standard treatment. In single-subject research, participants act as their own no-treatment controls and a participant's behavior during the baseline (no treatment) and treatment phases are compared.

Answer B: A cross-sectional design involves examining data from a population at one specific point in time.

Answer C: In ex post facto design, the researcher does not manipulate the independent variable as it has already occurred.

Answer D: In a double-blind study design, neither the researcher nor the participants know who is receiving a particular treatment. This is a between-groups design in which there is a control group for comparison.

Question 13

Not answered

Marked out of 1.00

Question ID #11723: Researchers would use an analysis of covariance (ANCOVA) to ____.

Select one:

- ☐ A. analyze the effects of an independent variable on two or more dependent variables
- ☐ B. statistically control variability in the dependent variable due to the effects of an extraneous variable
- ☐ C. adjust for systematic differences between groups that occurred as part of the experimental treatment
- ☐ D. determine if there is a significant interaction between two or more independent variables

The correct answer is B.

The ANCOVA is used to statistically remove the effects of an extraneous variable on the dependent variable. When using the ANCOVA, the extraneous variable is the "covariate" and its effects on the dependent variable are statistically removed so that it is easier to detect the effects of the independent variable on the dependent variable.

Answer A: This describes the MANOVA.

Answer C: Differences between groups as a result of experimental treatment is the desired outcome in experimental research and does not describe the ANCOVA.

Answer D: A factorial ANOVA is useful for determining if there is an interaction between independent variables. There is a factorial ANCOVA that can be used when a study includes two or more independent variables, but this question is asking about the ANCOVA more generally.

Question 14

Not answered

Marked out of 1.00

Question ID #11725: A psychologist conducts a t-test and concludes that the results are "significant at the .01 level." This means that ____.

Select one:

- ☐ A. there is a 1% chance the psychologist will incorrectly reject the null hypothesis
- ☐ B. there is a 1% chance the psychologist will incorrectly retain the null hypothesis
- ☐ C. there is a 99% chance the psychologist will incorrectly reject the null hypothesis
- ☐ D. there is a 99% chance the psychologist will incorrectly retain the null hypothesis

The correct answer is A.

The level of significance (alpha) determines the location of the boundary between the regions of likely and unlikely values in the sampling distribution. When results are significant at the chosen level of significance, this means that the results are in the region of unlikely values and that the null hypothesis should be rejected. Significance at the .01 level means that there is a 1% chance that the obtained value (e.g., the mean or the difference between means) could have occurred by chance alone given the value specified in the null hypothesis. In other words, there is a 1% probability that the null hypothesis will be incorrectly rejected or that a Type I error will be made.

Answer B: The probability of incorrectly retaining the null hypothesis (i.e., of retaining a false null hypothesis) is equal to beta, not alpha.

Answer C: There is a 1% probability, not 99%, that the researcher will incorrectly reject the null hypothesis, making a Type I error.

Answer D: Because the results are statistically significant, the psychologist will reject (not retain) the null hypothesis.

Question 15

Not answered

Marked out of 1.00

Question ID #11726: A researcher would use "counterbalancing" to ____.

Select one:

- ☐ A. control multiple treatment interference
- ☐ B. ensure an equal number of participants in each group
- ☐ C. match subjects on a confounding variable
- ☐ D. ensure that the sample is representative of the population from which it was drawn

The correct answer is A.

Counterbalancing involves administering the treatments (different levels of the IV) in different orders to different groups of participants. It helps control multiple treatment interference (also known as carry-over or order effects) that may result when multiple levels of the independent variable(s) are administered to the same participants.

Answer B: Counterbalancing involves administering the treatments in different orders to different groups of participants. It is not reliant on having equal number of participants in groups.

Answer C: Counterbalancing involves administering the treatments in different orders to different groups of participants. It is not related to matching subjects on confounding variables.

Answer D: This answer relates to sampling (random sampling, sample size, etc.) and not to treatment administration or counterbalancing.

Question 16

Not answered

Marked out of 1.00

Question ID #11727: "Sampling error" is ____.

Select one:

- ☐ A. the tendency of sample statistics to vary from one another due to unintended effects of the treatment
- ☐ B. the tendency of sample statistics to vary from population parameters due to the effects of systematic error
- ☐ C. the tendency of sample statistics to vary from population parameters due to the effects of random error
- ☐ D. the tendency of sample statistics to vary from population parameters due to the unreliability of the measuring instruments

The correct answer is C.

Sampling error is due to chance variations that can be expected in samples that have been randomly drawn from the population. Sampling error refers to the discrepancies between sample values and corresponding population values (parameters) that are due to chance factors in the selection process. Note that, by definition, sampling error is random, not systematic.

Answer A: The tendency of sample statistics to vary from one another due to unintended effects of the treatment addresses error in the form of extraneous and confounding variables.

Answer B: The variation in population parameters is due to random error, not a systematic error.

Answer D: Error due to the unreliability of the measuring instruments is referred to as measurement error, not sampling error.

Question 17

Not answered

Marked out of 1.00

Question ID #11728: To assess the effectiveness of a behavioral intervention on the caloric intake of young women with anorexia nervosa, you obtain a sample of women who have just received this diagnosis and determine the average number of calories each woman consumes each day during the week before and the week after the intervention is applied. Which of the following statistical tests will you use to analyze the collected data?

Select one:

- ☐ A. T-test for a single sample
- ☐ B. T-test for dependent (or correlated) samples
- ☐ C. T-test for independent samples
- ☐ D. T-test for multiple samples

The correct answer is B.

There are several versions of the t-test, but each is used to compare two means. The t-test for dependent (or correlated) samples is the appropriate test when the two means being compared are related, which is the case when they have been obtained from the same group of participants. Additional information on the t-test is provided in the Statistics and Research Design chapter of the written study materials.

Answer A: The t-test for a single sample is used to compare an obtained sample mean to a known population mean.

Answer C: The t-test for independent samples is used to compare the means from two unrelated groups.

Answer D: There is no "t-test for multiple samples." All t-tests are used to compare two means.

Question 18

Not answered

Marked out of 1.00

Question ID #11729: When using a "multiple-baseline" design, a researcher will ____.

Select one:

- ☐ A. apply and then withdraw the same treatment multiple times
- ☐ B. change the performance criterion gradually over the course of the study
- ☐ C. compare two or more treatments that are simultaneously applied to the same participant(s)
- ☐ D. sequentially apply a treatment to different behaviors or participants

The correct answer is D.

The multiple-baseline design is a type of single-subject design. When using the multiple-baseline design, a treatment is sequentially applied to different baselines (i.e., to different behaviors, to the same behavior in different settings, or to the same behavior performed by different individuals). Unlike a "reversal" design, the multiple-baseline design does not require the withdrawal of the treatment during the course of the study.

Answer A: This describes the reversal (withdrawal) design.

Answer B: This describes the changing criterion design.

Answer C: A multiple-baseline does not involve two or more treatments that are applied simultaneously, so this response does not describe the multiple-baseline design.

Question 19

Not answered

Marked out of 1.00

Question ID #11731: Which of the following bivariate correlation coefficients would you use to determine the degree of association between gender and reaction time (as measured in seconds)?

Select one:

- ☐ A. Spearman rho
- ☐ B. Point biserial
- ☐ C. Phi coefficient
- ☐ D. Pearson r

The correct answer is B.

Focus on the scale of measurement when choosing a correlation coefficient. The point-biserial correlation coefficient is appropriate when one variable is dichotomous (e.g., gender) and the other is measured on an interval or ratio scale (i.e., reaction time in seconds is measured on a ratio scale).

Answer A: The Spearman rho is used when both variables are measured in terms of ranks. Neither of the variables in this study are rank-ordered.

Answer C: The phi coefficient is appropriate when both variables are true dichotomies (e.g., undergraduate degree/no undergraduate degree and employed/unemployed). In this example, one of the variables is measured on a ratio scale (reaction time in seconds).

Answer D: The Pearson r is the appropriate correlation coefficient when both variables are continuous (measured on an interval or ratio scale). In this example, one of the variables is dichotomous (gender).

Question 20

Not answered

Marked out of 1.00

Question ID #11732: If a teacher adds 10 points to each score in a distribution of scores, this will ____.

Select one:

- ☐ A. affect the mean of the distribution but not its standard deviation or variance
- ☐ B. affect the mean and range of the distribution but not its standard deviation or variance
- ☐ C. affect the standard deviation and variance of the distribution but not its mean
- ☐ D. not affect the mean, range, standard deviation, or variance of the distribution

The correct answer is A.

For the exam, you want to be familiar with the effects of adding or subtracting a constant to each score in a distribution or multiplying or dividing each score by a constant. Adding a constant to each score in the distribution affects the distribution's mean (the new mean is the original mean plus the constant). However, adding a constant does not affect the variability of scores - i.e., it does not change the range, standard deviation, or variance.

Answer B: Adding a constant (i.e., 10 points to each score) does not affect the variability of scores. It does not change the range, standard deviation, or variance.

Answers C and D: Adding 10 points to each score will affect the mean, resulting in a larger mean, but it does not affect the variability. The range, standard deviation, or variance does not change.

Question 21

Not answered

Marked out of 1.00

Question ID #11736: A psychologist compares the effectiveness of three brief interventions for obsessive-compulsive disorder by randomly assigning adults who have received this diagnosis to one of the three interventions and measuring their symptoms prior to the beginning of treatment and one week, one month, and six months following treatment. Which of the following types of research design is the psychologist is using?

Select one:

- ☐ A. Between-groups
- ☐ B. Within-subjects
- ☐ C. Mixed
- ☐ D. Counterbalanced

The correct answer is C.

This study has two independent variables: type of treatment and time. In this study, the type of treatment is a between-groups variable because each participant is assigned to one of the three interventions. Time is a within-subjects variable because all participants are being evaluated on the dependent variable at four different times. Studies that have both between-groups and within-subjects variables are referred to as mixed designs.

Answer A: The type of treatment is a between-groups variable since each participant is assigned to one of the three interventions, but a between-groups design only accounts for half of the independent variables in this study. Time is a within-subjects variable since all participants are being evaluated on the dependent variable at four different times. Because the psychologist used both between-groups and within-subjects independent variables, this is a mixed design.

Answer B: Time is a within-subjects variable since all participants are being evaluated on the dependent variable at four different times, but a within-subjects design only accounts for half of the independent variables in this study. Type of treatment is a between-groups variable since each participant is assigned to one of the three interventions. Because the psychologist used both between-groups and within-subjects independent variables, this is a mixed design.

Answer D: A counterbalanced design is a type of within-subjects design that involves administering the different levels of the independent variable to different groups of participants in a different order.

Question 22

Not answered

Marked out of 1.00

Question ID #11737: The scattergram for scores on a measure of test anxiety and a measure of math achievement reveals that students with moderate levels of anxiety obtain the highest achievement scores, while students with low and high levels of anxiety obtain lower achievement scores. If the Pearson r is used to measure the degree of the association between test anxiety and math achievement scores, the resulting correlation coefficient will ____.

Select one:

- ☐ A. underestimate the relationship between the variables
- ☐ B. overestimate the relationship between the variables
- ☐ C. under or overestimate the relationship between the variables
- ☐ D. accurately estimate the relationship between the variables

The correct answer is A.

To identify the correct answer to this question, you have to recognize that the question is describing a nonlinear relationship between anxiety and math achievement and know that an assumption underlying the use of the Pearson r is that the relationship between variables is linear. In this situation, the relationship between the variables is nonlinear (i.e., the assumption of linearity has been violated). As a result, the Pearson r will underestimate the relationship between test anxiety scores and math achievement scores.

Answer B: The relationship between the variables is nonlinear, so the assumption of linearity has been violated. As a result, the Pearson r will underestimate the relationship between test anxiety scores and math achievement scores, not overestimate the scores.

Answer C: the relationship between variables would be underestimated, not overestimated.

Answer D: If the assumption of linearity is violated, we cannot accurately estimate the relationship between the variables.

Question 23

Not answered

Marked out of 1.00

Question ID #11738: Which of the following behavioral observation techniques would be most useful for evaluating the extent to which a student remains on-task while the teacher is talking?

Select one:

- ☐ A. Event recording
- ☐ B. Latency recording
- ☐ C. Interval recording
- ☐ D. Duration recording

The correct answer is C.

Interval recording is useful for behaviors that do not have a clear beginning or end (e.g., on-task behavior) and involves recording whether or not the behavior occurred during each interval.

Answer A: Event recording (sampling) is useful for behaviors that occur infrequently, have a long duration, and/or leave a permanent record or other product.

Answer B: Latency recording is used to determine how long it takes for a behavior to begin after a specific antecedent event has occurred (e.g., after a verbal demand or request has been made).

Answer D: Duration recording is used to capture how long a behavior lasts and is most useful for behaviors with a clear beginning and end.

Question 24

Not answered

Marked out of 1.00

Question ID #11739: Trend analysis is a type of analysis of variance used when ____.

Select one:

- ☐ A. the independent variable is quantitative
- ☐ B. the dependent variable is qualitative
- ☐ C. a researcher uses a cross-sectional research design
- ☐ D. a researcher uses a factorial research design

The correct answer is A.

Trend analysis is a type of analysis of variance that is used to evaluate linear and nonlinear trends. Trend analysis is used when a study involves a quantitative independent variable and the researcher wants to determine the shape of the relationship between the independent and dependent variables. The results of the trend analysis indicate whether or not there is a statistically significant linear or nonlinear (quadratic, cubic, quartic) trend.

Answer B: Trend analysis is used for examining quantitative data, not qualitative data.

Answer C: A cross-sectional design is an observational study design and is not directly related to using trend analysis.

Answer D: Factorial research designs have more than one independent variable and are not related to trend analysis.

Question 25

Not answered

Marked out of 1.00

Question ID #11740: In a scatterplot, the correlation between two variables is equal or close to zero when ____.

Select one:

- ☐ A. the range of Y scores at each value of X is equal to the total range of Y scores
- ☐ B. the range of Y scores at each value of X is about the same
- ☐ C. the range of Y scores is equal to the range of X scores
- ☐ D. the range of Y scores at each value of X is different

The correct answer is A.

The correlation for two variables is zero when scores on one variable are not related to scores on the other variable. This is the only answer of those given that describes a lack of correlation between X and Y. When the range of Y scores at each value of X equals the total range of Y scores, the scattergram for X and Y is rectangular in shape, indicating a correlation of (or near) zero.

Answer B: This describes homoscedasticity which is good as long as the range of Y scores at each value of X is not equal to the total range of Y scores.

Answer C: This condition is not a zero correlation between X and Y.

Answer D: This describes heteroscedasticity. Although heteroscedasticity can produce an inaccurate estimate of the degree of association between variables, it does not necessarily result in a correlation coefficient of zero.

Question 26

Not answered

Marked out of 1.00

Question ID #11741: To reduce the likelihood that experimenter expectancy will bias the results of a research study, you would use which of the following techniques?

Select one:

- ☐ A. Single-blind
- ☐ B. Double-blind
- ☐ C. Random selection
- ☐ D. Random assignment

The correct answer is B.

When a researcher uses a double-blind procedure, research participants and the experimenter are "blind" to the experimental conditions (i.e., they do not know what group participants are in). Because the researcher does not know whether individual participants are in the experimental or control group, this reduces the likelihood of providing cues to participants that could bias the results of the study.

Answer A: In a single-blind study, the participants do not know what group they are in. Using this procedure would not eliminate experimenter expectancies because the experimenter would know which participants belonged to each group.

Answer C: Random selection is a procedure for choosing participants from the population you want to study and would not eliminate experimenter expectancy. To eliminate experimenter expectancy, you would want the experimenter to be blind or unaware of which treatment condition the participant was in.

Answer D: Random assignment of participants to groups would not eliminate experimenter expectancy because random assignment is used to assign participants to groups. To reduce the likelihood of experimenter expectancy, you would want the experimenter to be blind or unaware of which treatment condition the participant was in.

Question 27

Not answered

Marked out of 1.00

Question ID #11742: When a distribution of scores is normally shaped, approximately what percent of scores fall between the mean and the score that is one standard deviation above the mean?

Select one:

- ☐ A. 25
- ☐ B. 34
- ☐ C. 50
- ☐ D. 68

The correct answer is B.

To identify the correct answer to this question, you have to be familiar with the areas under the normal curve. When a distribution is normally shaped, approximately 34% of scores fall between the mean and the score that is one standard deviation above the mean.

Answers A and C: When a distribution is normally shaped, approximately 34% of scores fall between the mean and the score that is one standard deviation above the mean.

Answer D: In a normal distribution, approximately 68% of scores fall between the scores that are one standard deviation below and above the mean.

Question 28

Not answered

Marked out of 1.00

Question ID #11743: To increase statistical power, you would ____.

Select one:

- ☐ A. increase beta from .001 to .01
- ☐ B. increase alpha from .01 to .05
- ☐ C. use a nonparametric statistical test
- ☐ D. decrease the sample size

The correct answer is B.

The ability to reject the null hypothesis is increased by increasing alpha. When the null hypothesis is actually false, increasing the size of alpha also increases power. Power refers to the ability to detect (reject) a false null hypothesis. Factors that affect the size of power are described in the Statistics and Research Design chapter of the written study materials.

Answer A: Beta is the probability of retaining a false null hypothesis (i.e., making a Type II error), and the larger the size of beta, the greater the probability of retaining (not rejecting) the false null hypothesis. Researchers do not have direct control over the magnitude of beta.

Answer C: Parametric tests are more powerful than are nonparametric tests.

Answer D: Decreasing the sample size would decrease power.

Question 29

Not answered

Marked out of 1.00

Question ID #11749: To test the validity of a model that describes the causal relationships among a set of variables, you would use which of the following techniques?

Select one:

- ☐ A. Path analysis
- ☐ B. Logistic regression
- ☐ C. Cluster analysis
- ☐ D. Multitrait-multimethod matrix

The correct answer is A.

Path analysis and other structural equation modeling techniques are used to determine if sample data fit a predefined model that indicates the direction of the causal links between variables.

Answer B: Logistic regression is a multivariate technique used to identify the combination of predictors that most accurately categorizes individuals into criterion groups, and is not used to describe a causal relationship.

Answer C: Cluster analysis is used to "cluster" or categorize individuals or items. Causal relationships cannot be derived from this type of analysis.

Answer D: The multitrait-multimethod matrix is used to evaluate the construct validity of a measure and includes information about the test's convergent and divergent validity. This is not used to determine causal relationships among a set of variables.

Question 30

Not answered

Marked out of 1.00

Question ID #11750: A teacher converts students' raw scores on a 200-item test to percent correct scores. What scale of measurement do the converted scores represent?

Select one:

- ☐ A. Ordinal
- ☐ B. Interval
- ☐ C. Ratio
- ☐ D. Nominal

The correct answer is C.

Percent correct scores represent a ratio scale of measurement (i.e., they have the properties of order, equal intervals, and an absolute 0 point). With percent correct scores, it is possible to conclude that someone who obtains a score of 50% got twice as many items correct as someone whose score is 25%.

Answer A: Ordinal scale data are ordered categories where the distances between categories are not known.

Answer B: On an interval scale, there is order and the distance between two values is meaningful but there is no natural zero point.

Answer D: This scale describes variables that are categorized without having a natural rank or order.

Question 31

Not answered

Marked out of 1.00

Question ID #11753: An educational psychologist conducts a study to assess the effects of an educational program designed to improve the academic performance of children diagnosed with a mild intellectual disability. The psychologist matches 30 children with mild intellectual disability on the basis of their IQ test scores and randomly assigns the members of each matched pair so one member is assigned to the experimental group and the other is assigned to the control group. The psychologist administers the program to the children in the experimental group and following completion of the program, administers an achievement test to all children and calculates a total correct score for each child. Which of the following should the psychologist use to assess the effects of the treatment on achievement test scores?

Select one:

- ☐ A. T-test for a single sample
- ☐ B. T-test for independent (unrelated) samples
- ☐ C. T-test for dependent (related) samples
- ☐ D. Two-way ANOVA

The correct answer is C.

The t-test for dependent (related) samples is used when the data are collected from two correlated groups, which is the case when a researcher uses matching to assign participants to groups. This study includes one independent variable that has two levels (program and no program); the data to be analyzed represent a ratio scale of measurement (total number of correct items); matching was used to control an extraneous variable (IQ score).

Answer A: The t-test for a single sample is used to compare the mean from a single sample to a known population mean.

Answer B: The t-test for independent samples is used to compare the means of two unrelated samples. The study in this question involves comparing means obtained from dependent (related) samples.

Answer D: A two-way ANOVA is used when a study involves two independent variables. This study has one independent variable (educational program) with two levels.

Question 32

Not answered

Marked out of 1.00

Question ID #11754: An organizational psychologist consulting for a manufacturing company is developing a selection battery to help the company answer the following question: "What group (assembly-line workers or clerical workers) does the job applicant most closely resemble?" Which of the following multivariate techniques would be useful for this purpose?

Select one:

- ☐ A. Discriminant analysis
- ☐ B. Multiple regression
- ☐ C. Factor analysis
- ☐ D. Path analysis

The correct answer is A.

A discriminant analysis (also known as discriminant function analysis) involves using scores on two or more predictors to predict an individual's membership in a criterion group (i.e., it is used when the criterion is measured on a nominal scale). When the criterion (Y variable) is group membership, discriminant analysis is an appropriate method.

Answer B: Multiple regression is used to combine scores on several predictors to predict or estimate performance or status on a criterion that is measured on an interval or ratio scale. The criterion in this example is on a nominal scale.

Answer C: Factor analysis is used to determine the underlying factors (attributes) that are assessed by a set of test items, subtests, or other measures.

Answer D: Path analysis is used to test a model of the causal links among a set of variables. The psychologist in the example is not making causal links, rather they are attempting to determine which group the job applicant most closely resembles.

Question 33

Not answered

Marked out of 1.00

Question ID #11757: When designing a research study, an investigator would be concerned about the impact of pretest sensitization on the study's external validity if ____.

Select one:

- ☐ A. the investigator's knowledge of how participants performed on the pretest affects how the investigator scores the post-tests
- ☐ B. the participants know which group (experimental or control) the investigator assigned them to
- ☐ C. the investigator revises the post-test on the basis of how participants responded to the pretest
- ☐ D. taking the pretest changes how the participants react to the treatment

The correct answer is D.

Pretest sensitization occurs when the administration of a pretest "sensitizes" participants to the purpose of the research study and, as a result, alters their reaction to the treatment. This is a potential threat to external validity and the results of the study cannot be generalized to situations in which a pretest is not administered prior to delivery of the treatment.

Answer A: Internal validity is threatened when subjects' pretest and post-test scores are affected by the investigator's knowledge of test performance rather than the independent variable.

Answer B: Having prior knowledge of which group participants' have been assigned could affect the outcome which is unrelated to the independent variable.

Answer C: Instrumentation is a threat to internal validity.

Question 34

Not answered

Marked out of 1.00

Question ID #11758: The Central Limit Theorem predicts that the sampling distribution of means increasingly approaches ____.

Select one:

- ☐ A. a normal shape as the sample size increases regardless of the shape of the distribution of scores in the population
- ☐ B. a normal shape as the number of samples drawn from the population increases regardless of the shape of the distribution of scores in the population
- ☐ C. the shape of the population distribution as the size of the sample increases
- ☐ D. the shape of the population distribution as the number of samples drawn from the population increases

The correct answer is A.

The Central Limit Theorem makes three predictions about the sampling distribution of means: (1) if repeated random samples of size N are drawn from the population, as N increases in size, the sampling distribution of means increasingly approaches normal regardless of the shape of the population distribution; (2) the mean of the sampling distribution of means equals the population mean; and (3) the standard deviation of the sampling distribution of means equals the population standard deviation divided by the square root of N . This response is consistent with the first prediction of the Central Limit Theorem.

Answer B: The Central Limit Theorem assumes an infinite number of samples. In other words, it is the size of the samples, not the number of samples, that impacts the shape of the sampling distribution.

Answer C: This is not what is predicted by the Central Limit Theorem. Instead, the Central Limit Theorem predicts that, even when the population distribution is not normal, the sampling distribution of means will increasingly approach normal as the sample size increases.

Answer D: This is not what is predicted by the Central Limit Theorem.

Question 35

Not answered

Marked out of 1.00

Question ID #11760: Statistical regression is a potential threat to a study's internal validity when ____.

Select one:

- ☐ A. there is a statistically different number of participants in the study's treatment groups
- ☐ B. there is a statistically significant difference in the number of participants who drop out of the different treatment groups during the course of the study
- ☐ C. participants are selected for inclusion in the study because of their extremely high or low scores on the pretest
- ☐ D. the researcher is unable to randomly assign participants to the different treatment groups

The correct answer is C.

When participants are chosen for inclusion in a study because of their extreme scores on the pretest, it is likely that their scores will "regress to the mean" on the post-test, regardless of the effects of the treatment. This can make it difficult to ascertain the effects of the independent variable on the dependent variable. Statistical regression refers to the tendency of extreme scores to "regress to the mean" on retesting (i.e., for examinees who obtained extremely high or extremely low scores on a measure to obtain scores closer to the mean when retested).

Answer A: Unequal numbers of participants in treatment groups is not related to statistical regression and will not have an impact on an individual's scores regressing to the mean.

Answer B: Attrition (mortality) threatens internal validity when the characteristics of those who drop out differ in an important way from those who remained and does not impact an individual's scores regressing to the mean.

Answer D: The lack of random assignment does impact internal validity, but it is not related to statistical regression.

Question 36

Not answered

Marked out of 1.00

Question ID #11761: When using stepwise multiple regression, the addition of predictors to the equation is usually based on ____.

Select one:

- ☐ A. a predefined model or theory
- ☐ B. the magnitude of R-squared
- ☐ C. the proportion of accurate versus inaccurate decisions made
- ☐ D. the magnitude of the eigenvalues

The correct answer is B.

The types of multiple regression include simple, hierarchical, and stepwise. As its name implies, stepwise regression involves adding (or subtracting) predictors to the multiple regression equation one at a time. When using stepwise multiple regression, each predictor is retained in the equation as long as it contributes significantly to the total amount of variability in the criterion that is explained by the combined predictors. The proportion of explained variability is measured by R-squared.

Answer A: This describes hierarchical multiple regression.

Answer C: This sounds like the basis for determining if predictors have incremental validity, which may be a consideration in multiple regression. However, this is not the best answer to this question.

Answer D: This refers to factor analysis.

Question 37

Not answered

Marked out of 1.00

Question ID #11762: A researcher calculated a Cohen's d of .50 for the data collected in a study comparing two brief treatments for generalized anxiety disorder. This means that ____.

Select one:

- ☐ A. there was a difference of one-half standard deviation between the means of the two groups
- ☐ B. there was a difference of one-fourth standard deviation between the means of the two groups
- ☐ C. the percent of variance accounted for by the difference in treatments was 50%
- ☐ D. the percent of variance accounted for by the difference in treatments was 25%

The correct answer is A.

Cohen's d is a measure of effect size. It indicates the difference between the means of two groups in terms of standard deviations. A Cohen's d of .50 indicates that one group obtained a mean that is one-half standard deviation higher than the mean obtained by the other group. Additional information about methods for calculating an effect size is provided in the Statistics and Research Design chapter of the written study materials.

Answer B: There is a difference of one-half standard deviation, not one-fourth.

Answers C and D: Cohen's d is a measure of the difference between two groups in terms of standard deviation units and is not related to the percent of variance accounted for.

Question 38

Not answered

Marked out of 1.00

Question ID #11763: LISREL (linear structural relations analysis) would be the appropriate method of analysis if your causal model ____.

Select one:

- ☐ A. includes measured (observed) attributes only
- ☐ B. includes latent traits only
- ☐ C. includes both measured (observed) attributes and latent traits
- ☐ D. includes both correlated and uncorrelated attributes

The correct answer is C.

LISREL, a structural equation (causal) modeling technique, is used to test causal hypotheses about relationships among measured variables and the latent traits those variables are believed to measure. In contrast to path analysis models, which predict the causal relationships among measured attributes only, LISREL models incorporate both measured attributes and latent traits. Additional information about LISREL and path analysis is provided in the Statistics and Research Design chapter of the written study materials.

Answers A, B, and D: LISREL (linear structural relations analysis) is appropriate for data that includes BOTH measured (observed) attributes and latent traits.

Question 39

Not answered

Marked out of 1.00

Question ID #11764: A factorial design ____.

Select one:

- ☐ A. always includes two or more dependent variables
- ☐ B. always includes two or more independent variables
- ☐ C. is used to control order effects
- ☐ D. is used to evaluate the effects of pretesting

The correct answer is B.

The term "factor" in factorial design refers to independent variables. A factorial research design is any design that includes two or more "factors" (independent variables).

Answer A: A Factorial design always includes two or more independent variables, not dependent variables.

Answer C: A factorial design does not control for order effects, but techniques like counterbalancing do.

Answer D: The effects of pretesting are related to issues of external validity and not to factorial design.

Question 40

Not answered

Marked out of 1.00

Question ID #11765: A researcher fears that the results of the study have been biased by aspects of the experimental situation that provided participants with cues about the responses or reactions that were expected of them. This researcher is concerned about which of the following?

Select one:

- ☐ A. Treatment diffusion
- ☐ B. Demand characteristics
- ☐ C. Confirmation bias
- ☐ D. Placebo effect

The correct answer is B.

This question describes a type of reactivity in which the behavior of participants is affected by cues (in the experimental setting) that indicate what behaviors are expected of them. Cues in the environment that inform participants about the purpose of the study or indicate what behavior is required of them are referred to as demand characteristics.

Answer A: Treatment diffusion (also known as treatment spillover) occurs when participants in one group inadvertently receive (or learn about) the intervention that has been applied to participants in another group.

Answer C: Confirmation bias is the tendency to pay attention to information that confirms one's current beliefs or preconceptions.

Answer D: The placebo effect refers to an improvement in physical or mental health that occurs in response to a "fake" treatment (e.g., a sugar pill), apparently because of the individual's expectation that the treatment will be effective.

Question 41

Not answered

Marked out of 1.00

Question ID #11767: The denominator term in the F-ratio is reduced in magnitude by ____.

Select one:

- ☐ A. decreasing within-group variability
- ☐ B. increasing within-group variability
- ☐ C. increasing between-group variability and decreasing within-group variability
- ☐ D. decreasing between-group variability and increasing within-group variability

The correct answer is A.

The mean square within (MSW) is the denominator of the F-ratio and, as its name implies, is a measure of within-group variability. Within-group variability is a measure of error; decreasing within-group variability decreases error and the magnitude of the denominator of the F-ratio. Additional information on the F-ratio is provided in the Statistics and Research Design chapter of the written study materials.

Answer B: This would increase the size of the denominator of the F-ratio.

Answer C: MSW is a measure of within-group variability and isn't affected by between-group variability.

Answer D: Decreasing between-group variability AND increasing within-group variability does not reduce the magnitude of the F ratio.

Question 42

Not answered

Marked out of 1.00

Question ID #11768: A researcher is conducting a study to evaluate the effects of a behavioral treatment for controlling severe, self-destructive behaviors. Which of the following single-subject designs would the researcher most likely use for ethical reasons?

Select one:

- ☐ A. Reversal
- ☐ B. Multiple sequential withdrawal
- ☐ C. Multiple baseline
- ☐ D. Latin square

The correct answer is C.

When evaluating a treatment to eliminate self-destructive behaviors, a researcher would NOT want to use a research design that requires the removal or withdrawal of a successful treatment during the course of the study. The multiple baseline design would be an acceptable design in this situation since it does not require the withdrawal of the treatment from a baseline once it has been applied to that baseline.

Answer A: Reversal designs (e.g., ABA, ABAB) require the withdrawal of the treatment during subsequent baseline (A) phases and would not be an appropriate design in this situation due to the danger involved in removing treatment for dangerous behaviors.

Answer B: When evaluating a treatment to eliminate self-destructive behaviors, a researcher would not want to use a research design that requires the removal or withdrawal of a successful treatment during the course of the study. This design requires withdrawal of the treatment and would not be appropriate.

Answer D: The Latin square design is not a single-subject design. It is a group design that is used to control order effects.

Question 43

Not answered

Marked out of 1.00

Question ID #11769: A psychologist uses a single-subject design to assess the impact of an educational intervention designed to increase on-task behavior. When conducting the study, the psychologist will measure the participant's on-task behavior ____.

Select one:

- ☐ A. once at the beginning and once at the end of the study
- ☐ B. once during the baseline phase and once during the treatment phase of the study
- ☐ C. immediately after and several weeks or months after the end of the study
- ☐ D. at regular intervals during the baseline and the treatment phases of the study

The correct answer is D.

This is the only answer choice that accurately describes the measurement of the dependent variable when using a single-subject research design. The various single-subject research designs share two characteristics: First, each design includes at least one baseline (no treatment) phase and one treatment phase; second, the dependent variable is measured at regular intervals during each phase.

Answers A, B, and C: These designs should include at least one baseline (no treatment) phase and one treatment phase; a dependent variable that is measured at regular intervals during each phase.

Question 44

Not answered

Marked out of 1.00

Question ID #11770: In the context of research design, external validity refers to the researcher's ability to ____.

Select one:

- ☐ A. manipulate the independent variable
- ☐ B. randomly assign participants to treatment groups
- ☐ C. determine whether there is a causal relationship between variables
- ☐ D. generalize the results of the study to other individuals, settings, and conditions

The correct answer is D.

External validity refers to the ability to generalize the results of a study to groups, settings, and conditions other than those used in the study. Random selection of participants from the population is one factor that helps ensure adequate external validity in terms of generalizability to other individuals.

Answer A: The ability to manipulate the independent variable has a direct impact on a study's internal validity, which refers to the ability to determine if the independent variable has affected the dependent variable.

Answer B: Random assignment is important for internal validity.

Answer C: The ability to determine if there is a causal relationship between variables depends on a study's internal (not external) validity.

Question 45

Not answered

Marked out of 1.00

Question ID #11771: A bivariate correlation coefficient of .60 indicates _____ times as much shared variability between variables as does a bivariate correlation coefficient of .30.

Select one:

- ☐ A. two
- ☐ B. three
- ☐ C. four
- ☐ D. six

The correct answer is C.

The correlation coefficient is squared to determine the amount of shared variability. A correlation coefficient of .60 indicates that 36% of variability is shared variability, while a correlation of .30 indicates that 9% of variability is shared variability. Since 36% is four times larger than 9%, this means that a coefficient of .60 indicates four times as much shared variability as does a coefficient of .30.

Answers A, B, and D: The correlation coefficient is squared to determine the amount of shared variability. Since 36% is four times larger than 9%, this means that a coefficient of .60 indicates four times as much shared variability as does a coefficient of .30.

Question 46

Not answered

Marked out of 1.00

Question ID #11772: A "normal" distribution is ____.

Select one:

- ☐ A. flat (rectangular)
- ☐ B. bell-shaped
- ☐ C. S-shaped
- ☐ D. bimodal

The correct answer is B.

A normal distribution of scores is symmetrical and bell-shaped. It is also known as the "bell curve".

Answer A: Flattened curves usually represent platykurtic distributions.

Answer C: S-shaped curves are not normal distribution curves.

Answer D: A bimodal distribution has two modes giving the appearance of having two "peaks" or "humps".

Question 47

Not answered

Marked out of 1.00

Question ID #11773: A researcher makes a Type II error when they ____.

Select one:

- ☐ A. reject the null hypothesis when it is true
- ☐ B. reject the null hypothesis when it is false
- ☐ C. retain the null hypothesis when it is true
- ☐ D. retain the null hypothesis when it is false

The correct answer is D.

In inferential statistics, there are two types of decision errors: (1) rejecting a true null hypothesis and (2) retaining a false null hypothesis. Retaining a false null hypothesis is referred to as a Type II error. The Statistics and Research Design chapter of the written study materials includes a decision outcome table that you might find useful for clarifying the decisions that are made about the null hypothesis.

Answer A: Rejecting a true null hypothesis is referred to as a Type I error.

Answer B: Rejecting a false null hypothesis is a correct decision.

Answer C: Retaining a true null hypothesis is a correct decision.

Question 48

Not answered

Marked out of 1.00

Question ID #11774: A researcher investigates the nature of client-therapist interactions during the initial therapy session by having psychology graduate students assume the role of either the therapist or therapy client and role-play the first session of therapy. The researcher's study is best described as which of the following?

Select one:

- ☐ A. Nonparametric
- ☐ B. Cross-sectional
- ☐ C. Causal comparative
- ☐ D. Analogue

The correct answer is D.

In clinical research, an analogue study is a study in which the conditions are in some way an approximation (i.e., analogue) of actual clinical practice.

Answer A: Nonparametric tests do not rely on any distribution or assumptions of normality and do not relate to a study in which the conditions are an approximation (analogue) of actual clinical practice.

Answer B: Cross-sectional designs involve comparing two or more population groups, who are similar, but differ on the variable of interest.

Answer C: The causal comparative design is appropriate for determining the relationships between independent and dependent variables after the action or event has occurred. This type of design does not describe conditions that are meant to approximate clinical practice.

Question 49

Not answered

Marked out of 1.00

Question ID #11775: Which of the following statistical tests would be most appropriate for analyzing the relationship between gender and Holland's six occupational themes?

Select one:

- ☐ A. Single-sample chi-square
- ☐ B. Multiple-sample chi-square
- ☐ C. One-way ANOVA
- ☐ D. Factorial ANOVA

The correct answer is B.

The study described in this question has one independent variable (gender) and one dependent variable (Holland's six occupational themes), and the dependent variable is measured on a nominal scale. The multiple-sample chi-square test is appropriate for studies that include two or more variables and the data being analyzed represent a nominal scale.

Answer A: The single-sample chi-square test is appropriate for descriptive studies that include a single nominal variable.

Answer C: The one-way ANOVA is appropriate for studies assessing the impact of a single independent variable on a single dependent variable that is measured on an interval or ratio scale.

Answer D: The factorial ANOVA is appropriate for studies assessing the impact of two or more independent variables on a single dependent variable that is measured on an interval or ratio scale.

Question 50

Not answered

Marked out of 1.00

Question ID #11777: The least squares criterion is used for ____.

Select one:

- ☐ A. isolating latent traits in a factor analysis
- ☐ B. predicting true scores from obtained scores
- ☐ C. estimating shrinkage when cross-validating a multiple regression equation
- ☐ D. identifying the line of best fit in a regression analysis

The correct answer is D.

For the licensing exam, you want to have the least squares criterion associated with regression analysis. The least squares criterion is used to locate the regression line in a scatterplot so the amount of error in prediction is minimized when using the regression line or its equation to predict criterion scores.

Answer A: Eigenvalues (how much of the observed variance the factor explains) and scree plots are most often used in isolating latent traits in a factor analysis.

Answer B: Predicting true scores from obtained scores describes classical test theory.

Answer C: There are two general formula categories for estimating shrinkage when cross-validating a multiple regression equation: estimators of the squared population multiple correlation coefficient and those of the squared population cross-validity coefficient.

Question 51

Not answered

Marked out of 1.00

Question ID #11778: In which of the following studies would an investigator use the split-plot ANOVA to analyze the collected data?

Select one:

- ☐ A. A study in which each participant receives only one level of Variable A and one level of Variable B
- ☐ B. A study in which participants who received level 1 of Variable A also received either level 1 or 2 of Variable B, while participants who received level 2 of Variable A received either level 3 or 4 of Variable B
- ☐ C. A study in which each participant received all levels of Variable A but in a different order
- ☐ D. A study in which each participant received only one level of Variable A but all levels of Variable B

The correct answer is D.

Split-plot is another name for "mixed," and a researcher would use a split-plot ANOVA when the study uses a mixed design (i.e., it includes both between-groups and within-subjects comparisons). This is the only answer that describes a mixed design: Variable A is a between-groups variable, while Variable B is a within-subjects variable. Additional information about between-groups and within-subjects designs is provided in the Statistics and Research Design chapter of the written study materials.

Answer A: This answer describes a study that includes two between-groups variables.

Answer B: This answer describes a nested design.

Answer C: This answer describes a counterbalanced design with a single independent variable.

Question 52

Not answered

Marked out of 1.00

Question ID #12247: An assumption underlying the use of regression analysis is that ____.

Select one:

- ☐ A. the relationship between X and Y is linear
- ☐ B. the relationship between X and Y is causal
- ☐ C. the X variables are uncorrelated
- ☐ D. the Y variables are highly correlated

The correct answer is A.

Another name for regression analysis is linear regression. Regression analysis is used to predict Y from a single X, and an assumption is that the relationship between X and Y is linear (i.e., the relationship can be described by a straight line). Additional information about regression analysis is provided in the Statistics and Research Design chapter of the written study materials.

Answer B: Regression analysis is NOT based on the assumption that the relationship between X and Y is causal.

Answer C: Regression analysis is the appropriate technique when there is a single X variable.

Answer D: Regression analysis is used when there is a single Y variable.

Question 53

Not answered

Marked out of 1.00

Question ID #12248: Which of the following sampling methods involves first selecting units of individuals rather than individuals and then either including all individuals from those units or a random sample of individuals from those units in the research study?

Select one:

- ☐ A. Stratified random sampling
- ☐ B. Cluster sampling
- ☐ C. Convenience sampling
- ☐ D. Systematic sampling

The correct answer is B.

Cluster sampling involves selecting clusters (units) rather than individuals. It is a useful technique when it is not feasible to identify or obtain access to individuals in the entire population. For example, rather than randomly selecting individual high school students from the population of all high school students, it may be necessary to select a sample of high schools and then include all students in those schools or a sample of students from those schools.

Answer A: Stratified random sampling involves first dividing the population into strata (segments) and randomly selecting individuals from those strata. Common strata include gender, SES, and race/ethnicity.

Answer C: A convenience sample involves selecting people on the basis of their availability.

Answer D: Systematic sampling is similar to simple random sampling and involves selecting every nth (e.g., 10th, 100th) participant from a list of individuals in the population beginning with the random selection of the first participant.

Question 54

Not answered

Marked out of 1.00

Question ID #12249: To statistically analyze rank-ordered data obtained from two correlated groups, you would use which of the following?

Select one:

- ☐ A. Mann-Whitney U
- ☐ B. Kruskal-Wallis
- ☐ C. Wilcoxon
- ☐ D. ANCOVA

The correct answer is C.

Knowing that the full name of this test is the Wilcoxon matched-pairs signed-ranks test would have helped you identify it as the correct response. The Wilcoxon is the appropriate test for rank-ordered data obtained from correlated groups (e.g., groups that have been formed by matching participants on the basis of their scores on a pretest or extraneous variable and then randomly assigning members of each matched pair to one of two groups). For the exam, you want to be familiar with the appropriate use of all of the tests listed in the answers to this question. Additional information on these tests is provided in the Statistics and Research Design chapter of the written study materials.

Answer A: The Mann-Whitney U test is used to analyze rank-ordered data from two independent groups.

Answer B: The Kruskal-Wallis test is used to analyze rank-ordered data from two or more independent groups.

Answer D: The ANCOVA is a type of analysis of variance that allows the researcher to statistically remove the effects of an extraneous variable. Like other forms of the analysis of variance, it is used when data represent an interval or ratio scale of measurement.

Question 55

Not answered

Marked out of 1.00

Question ID #12250: The analysis of variance produces an F-ratio, which is calculated by dividing MSB by MSW. The denominator of the F-ratio (MSW) is a measure of which of the following?

Select one:

- ☐ A. Treatment effects only
- ☐ B. Error only
- ☐ C. Treatment effects plus error
- ☐ D. Treatment effects minus error

The correct answer is B.

"Mean square within" is a measure of variability within treatment groups and provides an estimate of the effects of error. Knowing that MSB stands for "mean square between" may have helped you identify the correct answer to this question. (Additional information on the F-ratio is provided in the Statistics and Research Design chapter of the written study materials.)

Answer A: Since it is not possible to measure treatment effects only, they are estimated by dividing a measure of treatment plus error (MSB) by a measure of error only (MSW).

Answer C: "Mean square between" is the numerator of the F-ratio and provides a measure of variability due to a combination of treatment plus error.

Answer D: "Mean square between" is the numerator of the F-ratio and provides a measure of variability due to a combination of treatment plus error and "Mean square within" is in the denominator and provides an estimate of variability that is due to error.

Question 56

Not answered

Marked out of 1.00

Question ID #12252: Which of the following is the appropriate multivariate technique when using two or more continuous predictors to predict status on two or more continuous criteria?

Select one:

- ☐ A. Logistic regression
- ☐ B. Linear regression
- ☐ C. Discriminant function analysis
- ☐ D. Canonical correlation

The correct answer is D.

Of the techniques listed in the answers, canonical correlation is the only one that is appropriate when two or more continuous (interval or ratio) predictors will be used to predict status on two or more continuous criteria. For the exam, you want to be familiar with all four techniques listed in the answers to this question, which are described in the Statistics and Research Design chapter of the written study materials.

Answer A: Logistical regression is appropriate when the dependent variable is nominal and there is more than one independent variable.

Answer B: . Linear regression is used to examine the strength of the effect of the independent variable(s) on the dependent variable.

Answer C: Discriminant function analysis is used for predicting group membership base on a linear combination of interval predictor variables.

Question 57

Not answered

Marked out of 1.00

Question ID #12253: In the context of bivariate correlational techniques, "homoscedasticity" refers to ____.

Select one:

- ☐ A. a linear relationship between X and Y
- ☐ B. a nonlinear relationship between X and Y
- ☐ C. a different range of Y scores at different values of X
- ☐ D. the same range of Y scores at different values of X

The correct answer is D.

Homoscedasticity is one of the assumptions for using the Pearson r and most other correlation coefficients. This assumption is met when the range of Y scores is about the same at every value of X. Homoscedasticity and heteroscedasticity are illustrated in the Statistics and Research Design chapter of the written study materials.

Answer A: Having a linear relationship between X and Y is a condition for regression but does not describe homoscedasticity.

Answer B: A nonlinear relationship between X and Y does not meet the assumptions of regression and would therefore be the incorrect statistical technique.

Answer C: A different range of Y scores at different values of X is referred to as "heteroscedasticity".

Question 58

Not answered

Marked out of 1.00

Question ID #12254: When a distribution of scores is flatter than the normal distribution, we refer to it as ____.

Select one:

- ☐ A. platykurtic
- ☐ B. leptokurtic
- ☐ C. mesokurtic
- ☐ D. exokurtic

The correct answer is A.

"Platy" means flat or broad, and a platykurtic distribution is flatter than the normal distribution. For the exam, you want to be familiar with the normal curve and distributions that deviate from the normal curve. These are described in the Statistics and Research Design chapter of the written study materials.

Answer B: A leptokurtic distribution is more "peaked" (pointed) than a normal distribution.

Answer C: Mesokurtic is another term for a normal distribution.

Answer D: Exokurtic is a "made up" term.

Question 59

Not answered

Marked out of 1.00

Question ID #12255: A researcher assigns 100 high school students who are either high or low in math aptitude to one of the following groups: the traditional instruction, computer-assisted instruction, or combination (traditional plus computer-assisted instruction) group. After ten weeks of instruction, the researcher determines each student's score on a math achievement test. In this study, the independent variable(s) is/are ____.

Select one:

- ☐ A. type of instruction
- ☐ B. type of instruction and math aptitude
- ☐ C. math achievement test scores
- ☐ D. math aptitude and math achievement test scores

The correct answer is B.

This study is designed to answer the question, "What are the effects of math aptitude and type of instruction on math achievement test scores?" This study has two independent variables (type of instruction and math aptitude) and one dependent variable (math achievement test scores).

Answer A: Type of instruction is only one of two independent variables in this study, so this isn't the best answer.

Answer C: The math achievement test score is the outcome or dependent variable.

Answer D: Math aptitude is an independent variable, but math achievement test score is an outcome or dependent variable.

Question 60

Not answered

Marked out of 1.00

Question ID #12256: As a behavioral observation technique, event sampling (recording) is useful for behaviors that ____.

Select one:

- ☐ A. occur infrequently or have a long duration
- ☐ B. occur frequently or have a short duration
- ☐ C. occur randomly and involve a sequence of responses
- ☐ D. occur spontaneously and have no clear beginning or end

The correct answer is A.

Event sampling involves observing a behavior each time it occurs and is useful for behaviors that occur infrequently, have a long duration, or leave a permanent record. Methods of behavioral sampling include event sampling, interval recording, and situational sampling. These are described in the Statistics and Research Design chapter of the written study materials.

Answer B: This is the opposite of behaviors observed with event sampling, which are behaviors that occur infrequently or have a long duration.

Answer C: Event sampling does not rely on randomness or sequential events, but rather events that occur infrequently and/or have a long duration.

Answer D: Event sampling does not rely on behaviors with spontaneous occurrence or lacking a clear beginning or end, but rather events that occur infrequently and/or have a long duration.

Question 61

Not answered

Marked out of 1.00

Question ID #12257: An extraneous variable is a variable that is irrelevant to the purpose of a research study but confounds a study's results because it ____.

Select one:

- ☐ A. has a systematic effect on the study's dependent variable
- ☐ B. has a random effect on the study's dependent variable
- ☐ C. has a systematic effect on the study's independent variable
- ☐ D. has a random effect on the study's independent variable

The correct answer is A.

Knowing that an extraneous variable is a source of systematic error would have helped you narrow your choice of a correct response to answers a and c. An extraneous variable is a variable that is not relevant to the purpose of the study but has a systematic effect on the dependent variable and, as a result, can make it difficult to ascertain the effects of the independent variable on the dependent variable. Additional information on extraneous variables is provided in the Statistics and Research Design chapter of the written study materials.

Answer B: Confounding variables have a systematic effect (not random) on the dependent variables.

Answer C: Confounding variables have a systematic effect on the dependent variables and not the independent variables.

Answer D: This is the opposite of the correct answer. Confounding variables have a systematic effect on the dependent variables rather than random effects on independent variables.

Question 62

Not answered

Marked out of 1.00

Question ID #12337: An investigator is studying the impact of stress on the ability to perform a complex mental rotation task. The investigator is trying to eliminate all confounds to the experiment so they can conclude that increased stress causes decrements in processing speed. What type of validity is the investigator trying to establish?

Select one:

- ☐ A. Face validity
- ☐ B. Construct validity
- ☐ C. Internal validity
- ☐ D. External validity

The correct answer is C.

The investigator in this experiment is trying to establish a causal relationship between two variables - stress level and processing speed. Internal validity is the only type of validity that focuses on the causal relationship between independent and dependent variables.

Answer A: Face validity focuses on whether a test looks like it measures what it is intended to measure.

Answer B: Construct validity is established when a test measures the intended hypothetical trait.

Answer D: External validity focuses on the generalizability of one study to other conditions, individuals, etc.

Question 63

Not answered

Marked out of 1.00

Question ID #13807: Campbell and Stanley (1966) identify which of the following as a direct threat to a study's external, but not internal, validity?

Select one:

- ☐ A. Multiple treatment interference
- ☐ B. Statistical regression
- ☐ C. Maturation
- ☐ D. Attrition

The correct answer is A.

Anything that is a threat to a study's internal validity will also be a threat to its external validity. For this question, you need to identify a factor that is only a threat to external validity. Of the threats listed, only multiple treatment interference is listed by Campbell and Stanley as a direct threat to external validity. Multiple treatment interference occurs when a study involves exposing each subject to two or more levels of an independent variable (i.e., when the study utilizes a within-subjects design), the effects of one level of the independent variable can be affected by previous exposure to another level. When this occurs, the results of the study cannot be generalized to situations in which people will be exposed to only one level of the IV.

Answer B: Statistical regression is a threat to internal validity.

Answer C: Maturation is a threat to internal validity.

Answer D: Attrition is a threat to a study's internal validity.

Question 64

Not answered

Marked out of 1.00

Question ID #13905: When the number of participants in a research study is too small to use a completely counterbalanced research design, a researcher would use which of the following?

Select one:

- ☐ A. Changing criterion design
- ☐ B. Latin square design
- ☐ C. Multiple baseline design
- ☐ D. Solomon four-group design

The correct answer is B.

Counterbalancing is a within-subjects design that involves changing the order in which each treatment is administered to different groups of participants. The goal of counterbalancing is to use every possible sequence of treatments with an equal number of participants for each sequence. When the number of participants is too small, a researcher may use partial counterbalancing and the Latin square design is a type of partial counterbalancing that helps determine exactly what sequences of treatment will be administered to different groups of participants.

Answer A: The changing criterion design is a type of single-case design consisting of a series of phases in which a different behavioral criterion is set for each phase. If the behavior reaches the criterion level for each phase, the treatment is considered to be effective.

Answer C: The multiple baseline design is a type of single-case design that involves sequentially applying a treatment across participants, settings, or behaviors.

Answer D: The Solomon four-group design is used to evaluate the effects of pretesting on a study's internal and external validity.

Question 65

Not answered

Marked out of 1.00

Question ID #13906: A researcher would use which of the following techniques to evaluate the cause-and-effect or predictive relationships between measured variables and latent factors?

Select one:

- ☐ A. Meta-analysis
- ☐ B. Multitrait-multimethod matrix
- ☐ C. Discriminant function analysis
- ☐ D. Structural equation modeling

The correct answer is D.

Structural equation modeling is used to explore or confirm hypothesized relationships between both measured and latent variables.

Answer A: Meta-analysis is used to evaluate an intervention by combining the results of a number of research studies.

Answer B: The multitrait-multimethod matrix is used to evaluate convergent and divergent validity.

Answer C: Discriminant function analysis is used to classify people into criterion groups based on their scores or status on two or more predictors.

